ASSIGNMENT 1

Question 1: Check the pod in all namespaces with kubectl command and check the status of all the pods.

```
avej_lanjekar@AvejLanjekar:Assignment1$ kubectl get pods --all-namespaces
NAMESPACE         NAME                                        READY   STATUS
RESTARTS         AGE
cloudethix-avej   cloudethix-pod-9798fcf9d-5zsm9              1/1     Running
3                32h
cloudethix-avej   cloudethix-pod-9798fcf9d-t8wtq              1/1     Running
3                32h
default           cloudethix-static-minikube                  1/1     Running
3                29h
default           config-subpath                              1/1     Running
14               9d
default           database                                    1/1     Running
12               8d
default           mysql-0                                     1/1     Running
5                2d9h
default           mysql-1                                     1/1     Running
5                2d10h
default           order-service                               1/1     Running
15               9d
default           pod-a                                       1/1     Running
14               9d
default           pod-b                                       1/1     Running
14               9d
default           port-forward-pod                            1/1     Running
13               9d
default           pvc-test                                    1/1     Running
14               9d
default           sa-no-mount-test                            1/1     Running
14               9d
default           sa-test                                     1/1     Running
14               9d
default           svc-acc-pod                                 0/1
ContainerCreating   1 (9d ago)       9d
default           web-pod                                     1/1     Running
13               9d
ingress-nginx     ingress-nginx-controller-7d65c586d6-xx48q   1/1     Running
9                6d3h
kube-system       coredns-7d764666f9-wqx9g                    1/1     Running
```

```
14               9d
kube-system       csi-nfs-controller-7f8cc69c8-57d7x        5/5     Running
26               2d3h
kube-system       csi-nfs-node-5khhg                        3/3     Running
15               2d3h
kube-system       csi-nfs-node-glznq                        3/3     Running
15               2d3h
kube-system       csi-nfs-node-pd72g                        3/3     Running
9                29h
kube-system       etcd-minikube                             1/1     Running
14               9d
kube-system       kindnet-dkrww                             1/1     Running
3                29h
kube-system       kindnet-j4hzp                             1/1     Running
14               9d
kube-system       kindnet-wp88p                             1/1     Running
5                2d3h
kube-system       kube-apiserver-minikube                   1/1     Running
14               9d
kube-system       kube-controller-manager-minikube          1/1     Running
15               9d
kube-system       kube-proxy-g578q                          1/1     Running
63               2d3h
kube-system       kube-proxy-ss6r5                          0/1     Error
44 (2m22s ago)   29h
kube-system       kube-proxy-vbdw7                          1/1     Running
14               9d
kube-system       kube-scheduler-minikube                   1/1     Running
14               9d
kube-system       storage-provisioner                       1/1     Running
31 (4m8s ago)    9d
ns-1              backend-7cdc9fcc87-6fhgn                  1/1     Running
2                9h
ns-1              backend-7cdc9fcc87-w6ng5                  1/1     Running
2                9h
ns-1              backend-7cdc9fcc87-zhg5g                  1/1     Running
2                9h
ns-1              cloudethix-configmap-778f8bcdc7-fg5t9     1/1     Running
2                10h
ns-1              cloudethix-configmap-778f8bcdc7-fvlv9     1/1     Running
2                10h
ns-1              cloudethix-configmap-778f8bcdc7-htg2r     1/1     Running
2                10h
ns-1              config-inspector                          1/1     Running
3                35h
```

```
ns-1                config-subpath                              1/1     Running
3                 35h
```

Question 2: Create one nginx pod on your Kubernetes cluster and then delete the same.

Pod.YAML

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
    - name: nginx
      image: nginx
      ports:
        - containerPort: 80
```

```
avej_lanjekar@AvejLanjekar:Q2$ kubectl apply -f pod.yml
pod/nginx-pod created

avej_lanjekar@AvejLanjekar:Q2$ kubectl get pods
NAME                             READY   STATUS    RESTARTS   AGE
nginx-pod                        1/1     Running   0          5s

avej_lanjekar@AvejLanjekar:Q2$ kubectl delete -f pod.yml
pod "nginx-pod" deleted from default namespace
```

Question 3: Create a YAML file for ReplicationController and ReplicaSet. Update the name from metadata and add yourname_rc and yourname_rs to both objects respectively. Once files are ready, deploy the objects in your K8S cluster with kubectl command.

ReplicationController.yaml

```yaml
apiVersion: v1
kind: ReplicationController
metadata:
  name: avej-rc
spec:
  replicas: 2
  selector:
    app: nginx
  template:
    metadata:
      name: nginx
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

```
avej_lanjekar@AvejLanjekar:Q4$ ls
replicationController.yml

avej_lanjekar@AvejLanjekar:Q4$ kubectl apply -f replicationController.yml
replicationcontroller/avej-rc created

avej_lanjekar@AvejLanjekar:Q4$ kubectl get rc
NAME      DESIRED   CURRENT   READY   AGE
avej-rc   2         2         2       17s

avej_lanjekar@AvejLanjekar:Q4$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
avej-rc-7hztj                       1/1     Running   0          26s
avej-rc-lrxxk                       1/1     Running   0          26s
```

ReplicaSet.YAML

```yaml
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: avej-rs
spec:
  replicas: 2
  selector:
```

```yaml
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```

```
avej_lanjekar@AvejLanjekar:Q4$ kubectl apply -f replicaSet.yml
replicaset.apps/avej-rs created

avej_lanjekar@AvejLanjekar:Q4$ kubectl get rs
NAME                          DESIRED   CURRENT   READY   AGE
avej-rs                       2         2         2       5s
init-cont-deplo-db6fc8657     3         3         3       175m
node-example-89b8c475c        1         1         1       156m

avej_lanjekar@AvejLanjekar:Q4$ kubectl get pods
NAME                                 READY   STATUS    RESTARTS   AGE
avej-rc-7hztj                        1/1     Running   0          4m42s
avej-rc-lrxxk                        1/1     Running   0          4m42s
avej-rs-6d72n                        1/1     Running   0          19s
avej-rs-fx5q8                        1/1     Running   0          19s
```

Question 4:

Task Requirements

Clone the repository:

`https://github.com/zembutsu/docker-sample-nginx.git`

- Initialize one repo on your local system named Cloudethix-nginix-yourname and then copy all the content from cloned repo to your local repo.
- Update the index.html and add the name CloudEthiX along with Hostname. Commit the change to your local repository. Then create and push the code from the local repository to your remote repository named Cloudethix-nginix-yourname under master branch.

- Once code is pushed, create a Docker hub public repository named Cloudethix-nginix-yourname.
- Then build the docker image out of the dockerfile. Once an image is built, tag the image and push that image to the docker hub repo you created.
- Create a deployment.yaml file with 2 replications in cloudethix_yourname Namespace and use the container image that you have pushed to your docker hub repo. Once a file is created, run the deployment in k8s cluster.

Updated index.html

```html
<html>
<body>
        <h1>Host:  CloudEthiX <!--#echo var="HOSTNAME" --></h1>
        Version: 1.1
</body>
</html>
```

```
avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ ls
Dockerfile  LICENSE  README.md  default.conf  index.html

avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ git init
Initialized empty Git repository in /home/avej_lanjekar/kubernetes-
Assignment/Repetition-1/Assignment1/Q5/Cloudethix-nginx-Avej/.git/

avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ git status
On branch master

No commits yet

Untracked files:
  Dockerfile
  LICENSE
  README.md
  default.conf
  index.html

avej_lanjekar@AvejLanjekar:~/kubernetes-Assignment/Repetition-
1/Assignment1/Q5/Cloudethix-nginx-Avej$ git add .

avej_lanjekar@AvejLanjekar:~/kubernetes-Assignment/Repetition-
1/Assignment1/Q5/Cloudethix-nginx-Avej$ git commit -m "Initial Commit , adding
all files"
[master (root-commit) 25d68aa] Initial Commit , adding all files
 5 files changed, 46 insertions(+)
```

```
 create mode 100644 Dockerfile
 create mode 100644 LICENSE
 create mode 100644 README.md
 create mode 100644 default.conf
 create mode 100644 index.html

avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ git status
On branch master
nothing to commit, working tree clean

avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ git remote add origin
git@github.com:Avejlanjekar/Cloudethix-nginix-AvejLan.git

avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ git branch -M main

avej_lanjekar@AvejLanjekar:~ Cloudethix-nginx-Avej$ git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.38 KiB | 202.00 KiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Avejlanjekar/Cloudethix-nginix-AvejLan.git
 * [new branch]      main -> main
```

Deployment.YAML

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: cloudethix-pod
  namespace: cloudethix-avej
spec:
  replicas: 2
  selector:
    matchLabels:
      app: cloudethix
  template:
    metadata:
      labels:
        app: cloudethix
    spec:
      containers:
        - name: cloudethix
          image: avejlanjekar45/cloudethix-nginix-avej
```

```yaml
        ports:
          - containerPort: 80
```

```
avej_lanjekar@AvejLanjekar:~$ kubectl get deployments -n cloudethix-avej
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
cloudethix-pod   2/2     2            2           4m16s

avej_lanjekar@AvejLanjekar:~ kubectl get pods -n cloudethix-avej
NAME                              READY   STATUS    RESTARTS   AGE
cloudethix-pod-9798fcf9d-5zsm9   1/1     Running   0          4m23s
cloudethix-pod-9798fcf9d-t8wtq   1/1     Running   0          4m23s
```
